AI & Law Summerschool 2026
Session 2.2.1:
Evidential reasoning

Henry Prakken

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Utrecht University

Uncertainty in legal proof in criminal cases

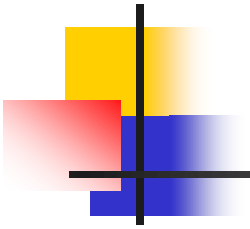
- Legal proof of facts is never completely certain
  - Eye witnesses can be unreliable
  - Expert witnesses sometimes disagree
  - Documents can be manipulated
  - Generalisations can have exceptions
  - DNA tests have an error margin
  - Confessions might be false
  - …



Models of proof that leave room for uncertainty

- Bayesian models
  - Apply Bayesian probability theory to determine how probable the charge is given the evidence
- Argumentation
  - Construct an argument for the statement to be proven from the available evidence
  - Defend this argument against all possible counterarguments
- Story-based models
  - Construct stories that explain the evidence
  - Choose the best
    - The most coherent, giving the best explanation of the evidence

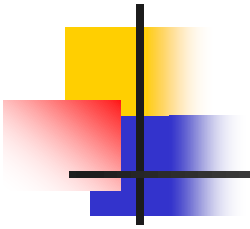
H. Prakken, F.J. Bex, A.R. Mackor & (eds), Models of Rational Proof in Criminal Law. *Topics in Cognitive Science* 12:4 (2020), special issue.



Question

- We want a model of rational proof that is:
  - rationally well-founded
  - cognitively feasible

AI support for legal proof: the KA bottleneck



- Required knowledge (common sense!) is hard to manually acquire and code
- And too diverse to automatically learn from natural-language sources
  - But what about GenAI?
- Some limited applications (e.g. BNs for frequent types of forensic evidence)
- Most AI applications are 'pedagogic' or for 'sense making'



Rational legal proof with argumentation

F.J. Bex, H. Prakken, C. Reed & D. Walton (2003), Towards a formal account of reasoning about evidence: argumentation schemes and generalisations. *Artificial Intelligence and Law* 11: (2003) 125-165.

Example: Adams

A rape took place in Hemel Hempstead in 1991. In 1993 Denis John Adams was arrested for another offence and a routine check showed that his DNA matched with that of a sample of semen obtained from the victim of the 1991 rape.

The prosecution's forensic expert estimated the random match probability as 1 in 200 million; the [illegible] in 2 million was a better estimate.

If you think he is, then you commit the 'prosecutor fallacy'

Do you think that Adams was guilty of the 1991 rape?

Example: Adams v Regina

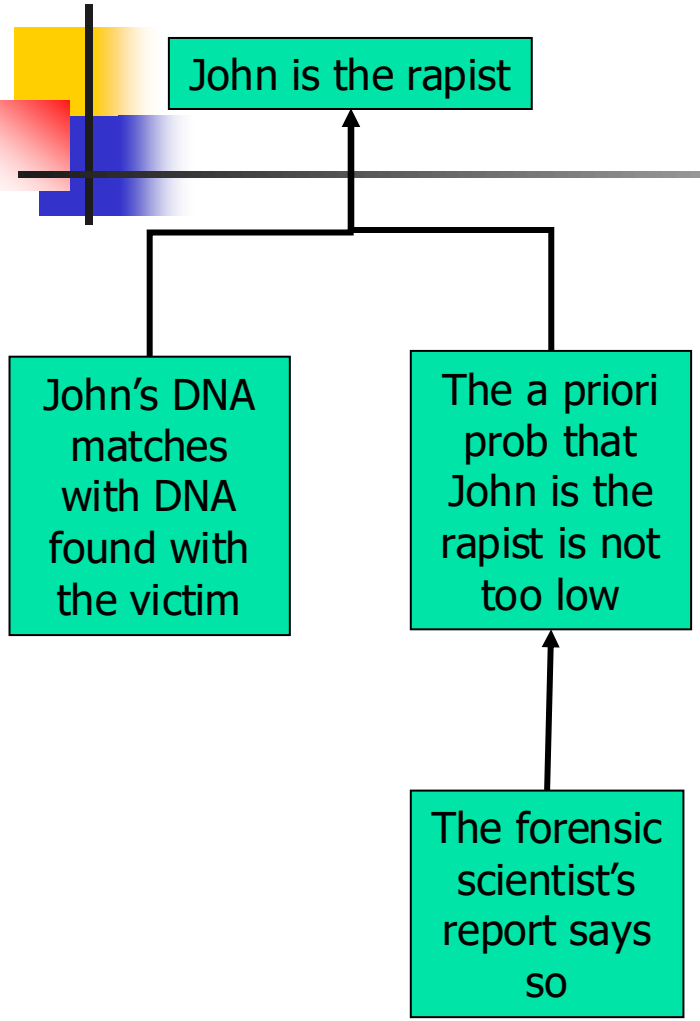
A rape took place near London (UK) in 1991. In 1993 Denis John Adams was arrested for another offence and a routine check showed that his DNA matched with that of a sample of semen obtained from the victim of the 1991 rape.

The prosecution's forensic expert estimated the random match probability as 1 in 200 million; the defence thought that 1 in 2 million was a better estimate.

A line up took place but the victim did not recognize Adams, and she said he did not resemble her attacker. Adams was 37 and looked older, while the victim claimed the rapist was in his early twenties.

Adams' girlfriend testified that he had spent the night of the attack with her.

Arguments and counterarguments



Regina v. Adams 1995



John is the rapist

John is not the rapist

John's DNA matches with DNA found with the victim

The a priori prob that John is the rapist is not too low

The forensic scientist's report says so

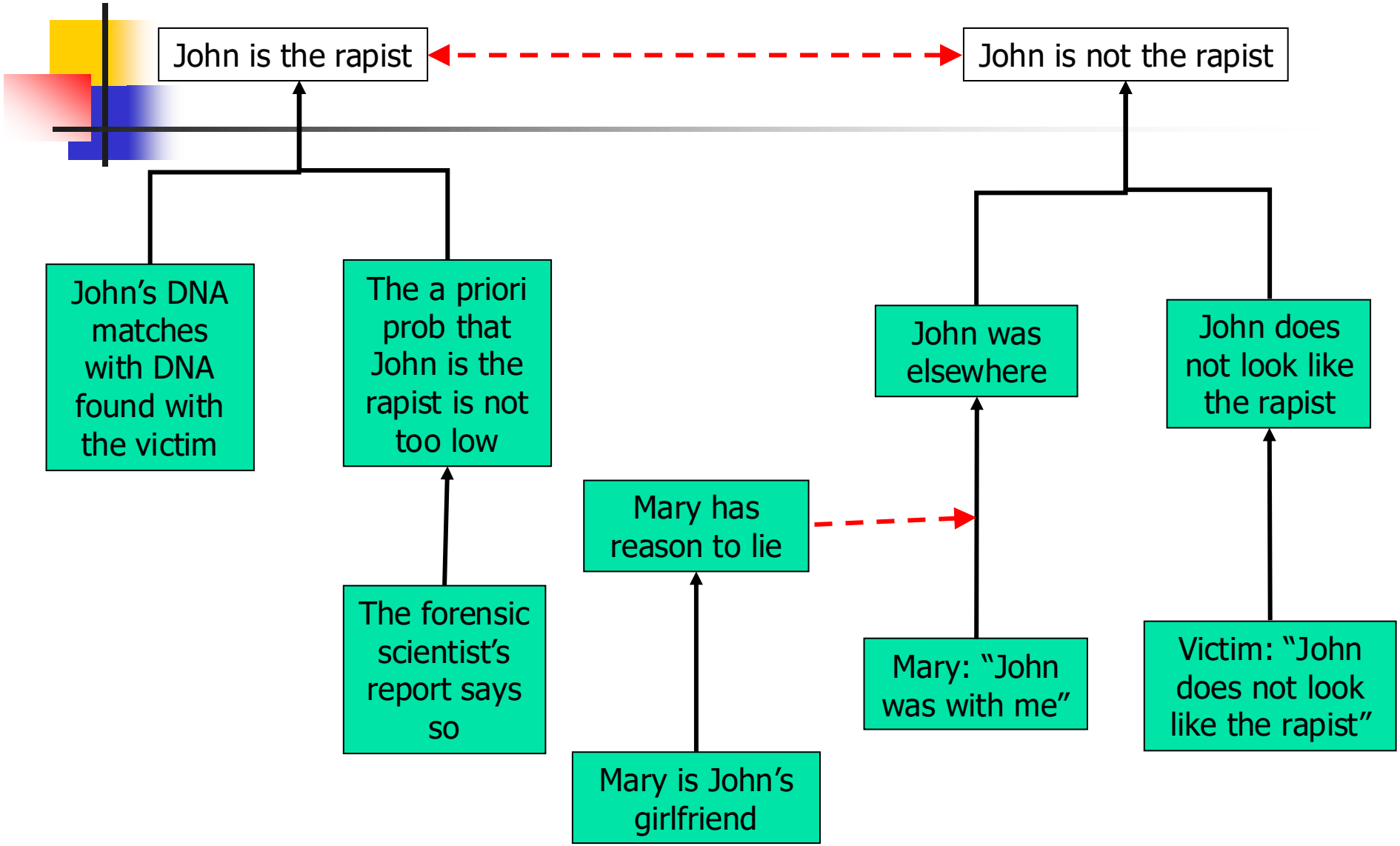
John was elsewhere

Mary: "John was with me"

John does not look like the rapist

Victim: "John does not look like the rapist"




John is the rapist
John is not the rapist
John's DNA matches with DNA found with the victim
The a priori prob that John is the rapist is not too low
The forensic scientist's report says so
Mary has reason to lie
Mary is John's girlfriend
John was elsewhere
Mary: "John was with me"
John does not look like the rapist
Victim: "John does not look like the rapist"

Argument scheme from Witness testimony



Witness W says "P"
Therefore (presumably), P

- Critical questions:
  - Does W have reason to lie?
  - Is W's memory OK?
  - Were W's senses OK?

Legal proof with arguments: critical questions

- Which generalisations are the glue in the arguments?
- Do they hold in general?
- If so: are there exceptions in this case?
  - Critical questions of argument schemes can help
- Can counterarguments (if any) be refuted?

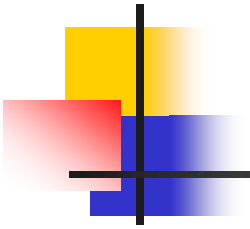


Rational legal proof with stories/scenarios

Pennington & Hastie (1993)

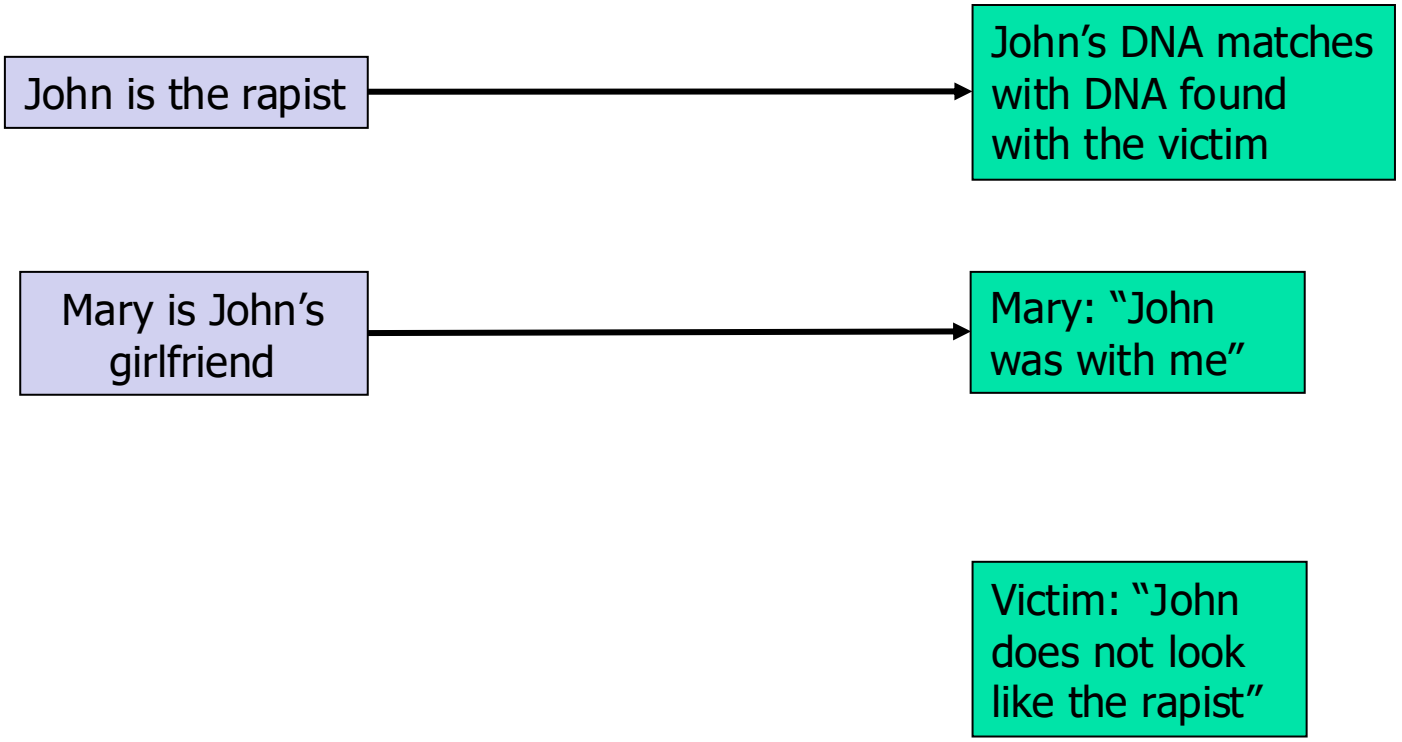
- Judges and jurors can only understand evidential reasoning in the form of stories about what may have happened.
- A good story:
  - Explains the evidence
  - Is coherent:
    - Is internally consistent
    - Is plausible: conforms to our general knowledge about the world
    - Is complete: contains initiating events which cause the main actor to have intentions, which give rise to actions, which have consequences

Argumentation vs. scenario approach

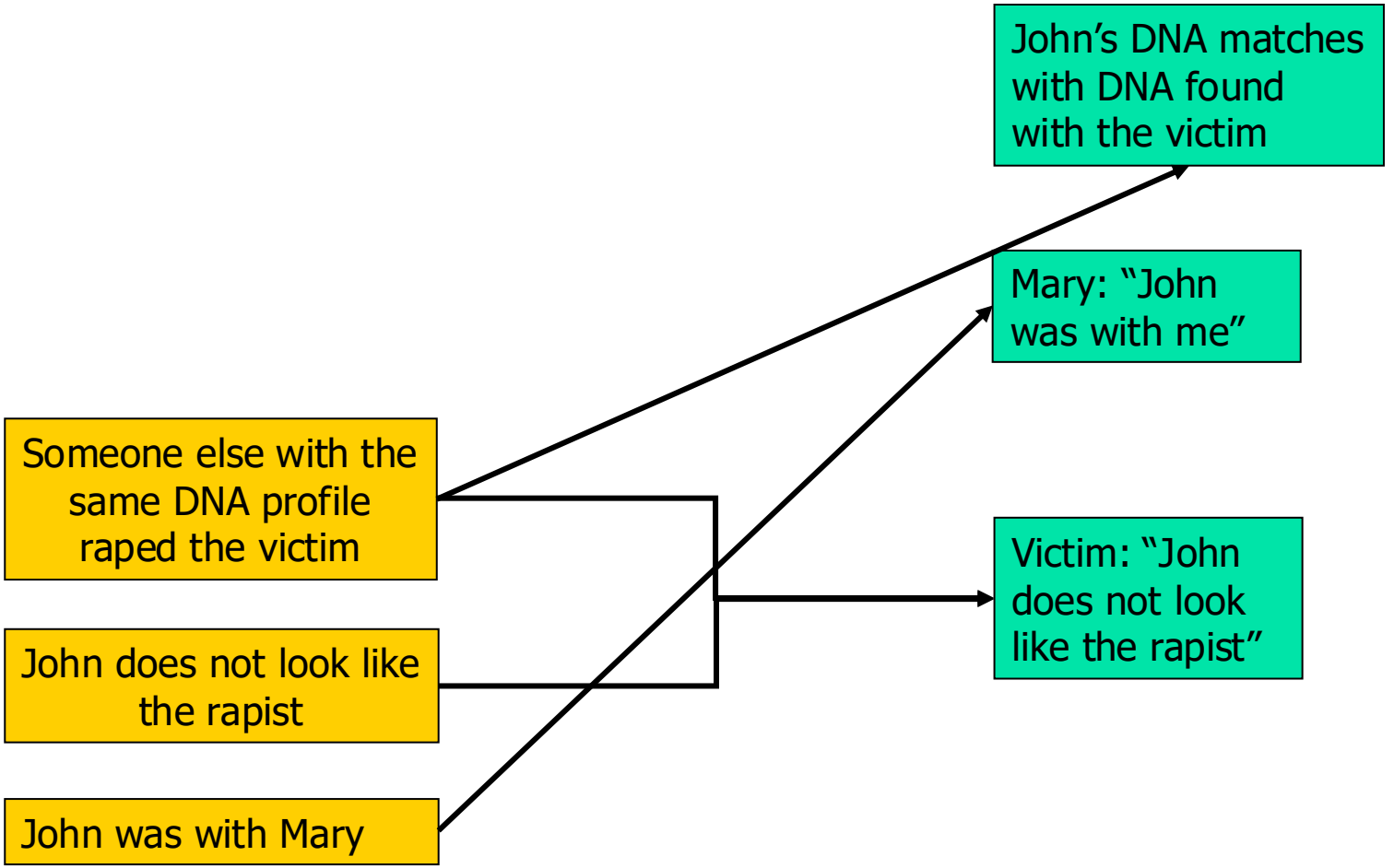


- Argumentation: reasons from evidence to hypotheses by applying evidential generalisations / argument schemes
  - Critical testing by searching for counterarguments based on exceptions
- Scenario approach: assumes the hypotheses and then tests whether the evidence (likely) follows by applying causal generalisations
  - Critical testing by checking under which hypothesis the evidence most likely follows

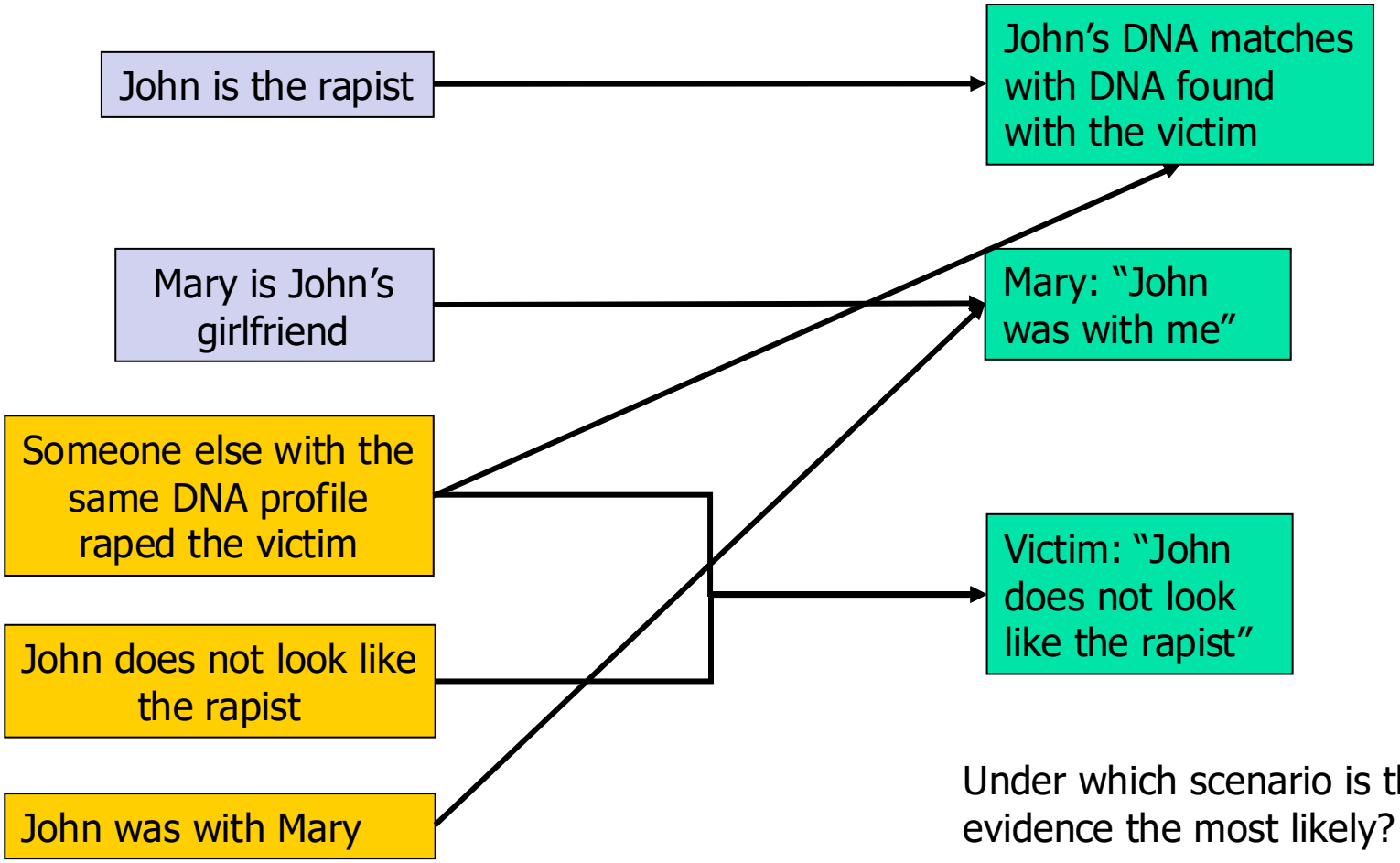
Constructing and comparing scenario's



Constructing and comparing scenario's

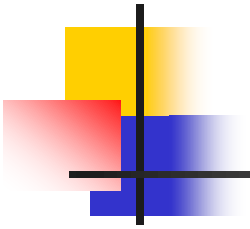


Constructing and comparing scenario's



Under which scenario is the evidence the most likely?

Legal proof with scenarios: critical questions

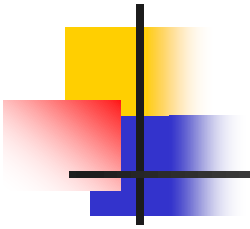


- Is the scenario plausible, consistent and complete?
- Does it explain the evidence?
- Are there alternative scenario's that explain the evidence?
- If so: how likely is the evidence given the various scenarios?
- …



Rational legal proof with Bayesian probability theory

H. Prakken, Recognising and avoiding fallacies in interpreting statistical evidence. To appear(?) in S. Brewer, J. Maranhao & G. Sartor (eds), *Logic and Law*. Edward Elgar Publishing. https://webspace.science.uu.nl/~prakk101/pubs/BayesLawLogic.pdf



Probability theory

- Every proposition has a probability between 0 and 1 (or between 0% and 100%)
  - The probability of A is 1 (100%) means: A is certainly true
  - The probability of A is 0 (0%) means: A is certainly false
- When A and B are each other's opposites, their probabilities add up 1 (100%)
  - The probability of A is 1 (100%) minus the probability of not-A
- Also holds for conditional probabilities: the probability of A given B
  - The probability of A given B = 1 (100%) minus the probability of not-A given B

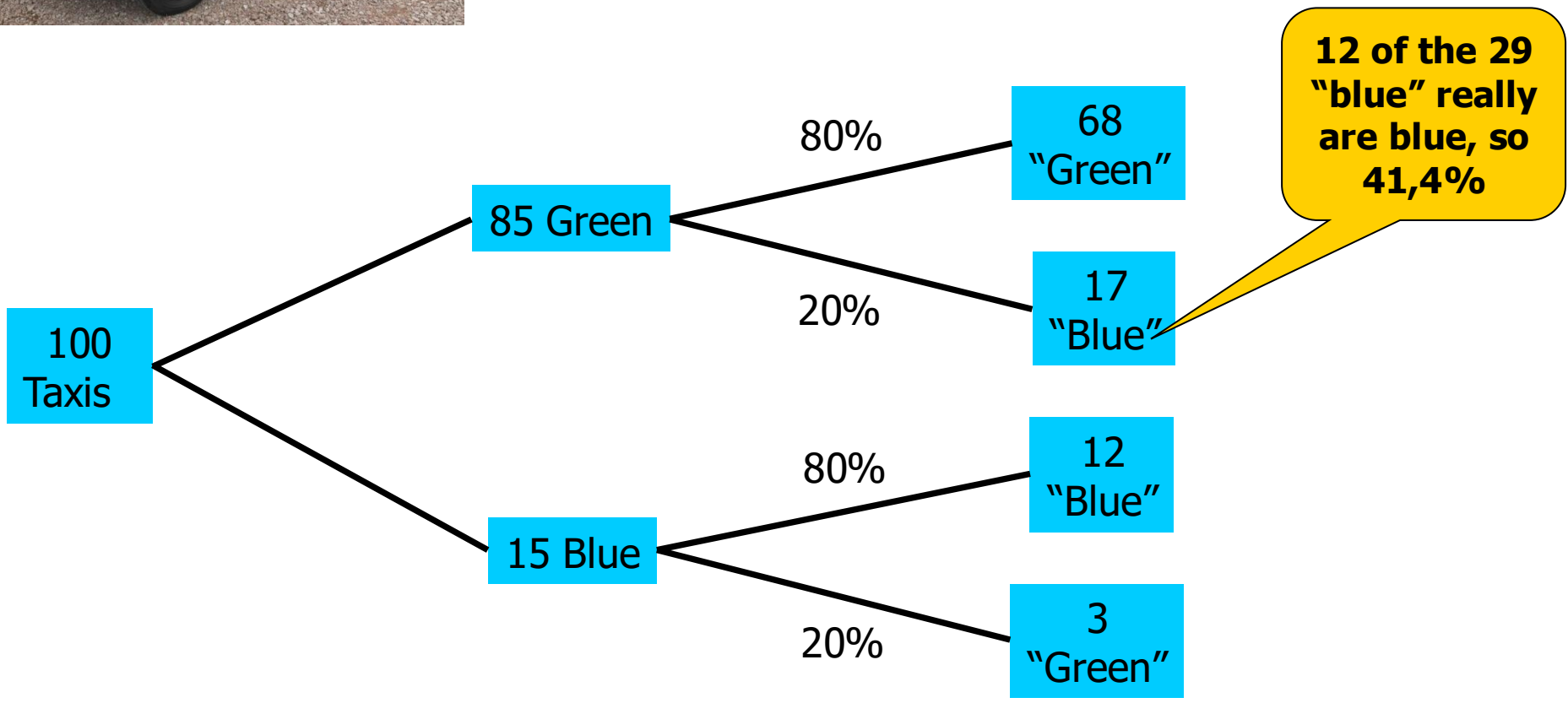
Example: blue and green taxis

On a misty winter night a taxi in a certain town sideswipes another car and drives off. A witness says that it was a blue taxi.
85% of the taxis in the town are owned by company Green Cabs Ltd and are all green, while the remaining 15% is owned by Blue Taxi Inc and are all blue.
The witness is tested and in 80% of the cases she correctly reports the colour of the car that is seen.

What do you think is the probability that the car that hit the other car is of Blue Taxi Inc, that is, the probability that the car is Blue given that the witness says "the car is Blue"?



85 green and 15 blue taxis
Witness is 80% reliable
Witness: "Blue"



Fatherhood test example: www.dnavaderschapstest.nl

"In a legally valid DNA paternity test, Verilabs shows with a certainty of more than 99.99% whether a man is the biological father of a child"

Verilabs test whether John is the father of Mary's child. The test is positive.

Is John is the father of Mary's child?

Fatherhood test example: www.dnavaderschapstest.nl

"In a legally valid DNA paternity test, Verilabs shows with a certainty of more than 99.99% whether a man is the biological father of a child"

Verilabs test whether John is the father of Mary's child. The test is positive.

Which probability is "more than 99,99%"?

It is not:

The probability that John is the father given a positive test, and that he is not the father given a negative test.

It is:

The probability of a positive test given that John is the father, and of a negative test given that he is not the father.

Fatherhood test in tree form

The probability of a negative test given that the person is not the father, is ~~more than~~ 99,99%

So:

The probability of a positive test given that the person is not the father, is 0,01% = 1 in 10.000

A priori

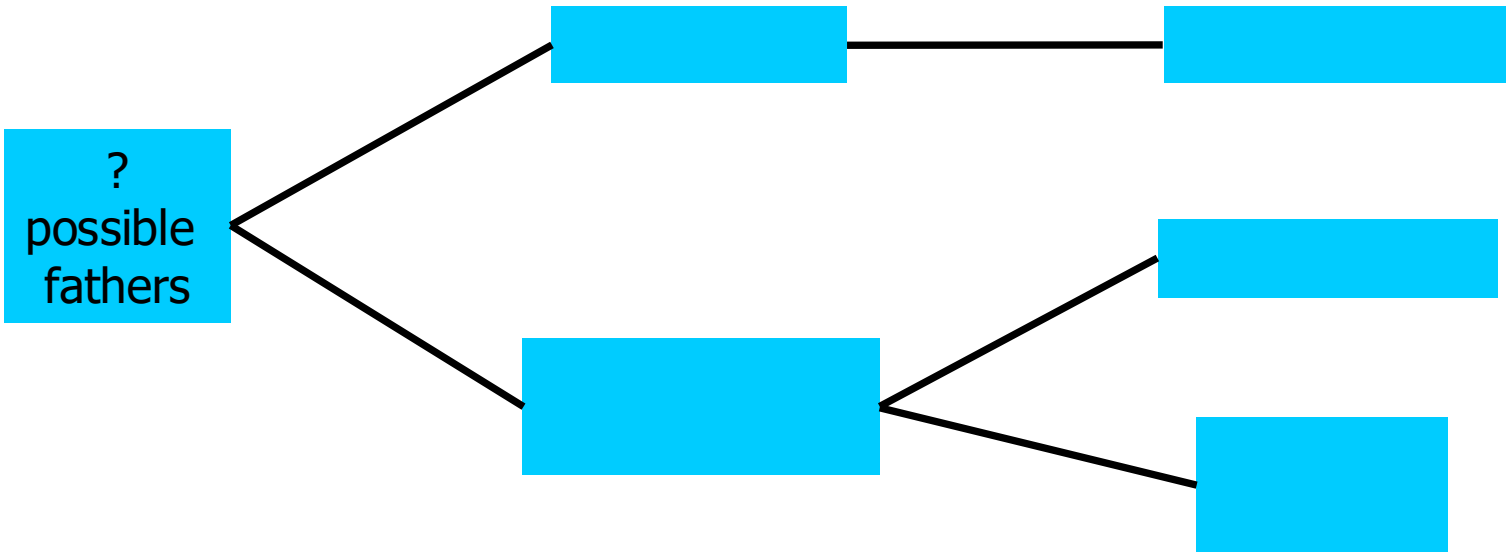
Evidence

Fatherhood test in tree form

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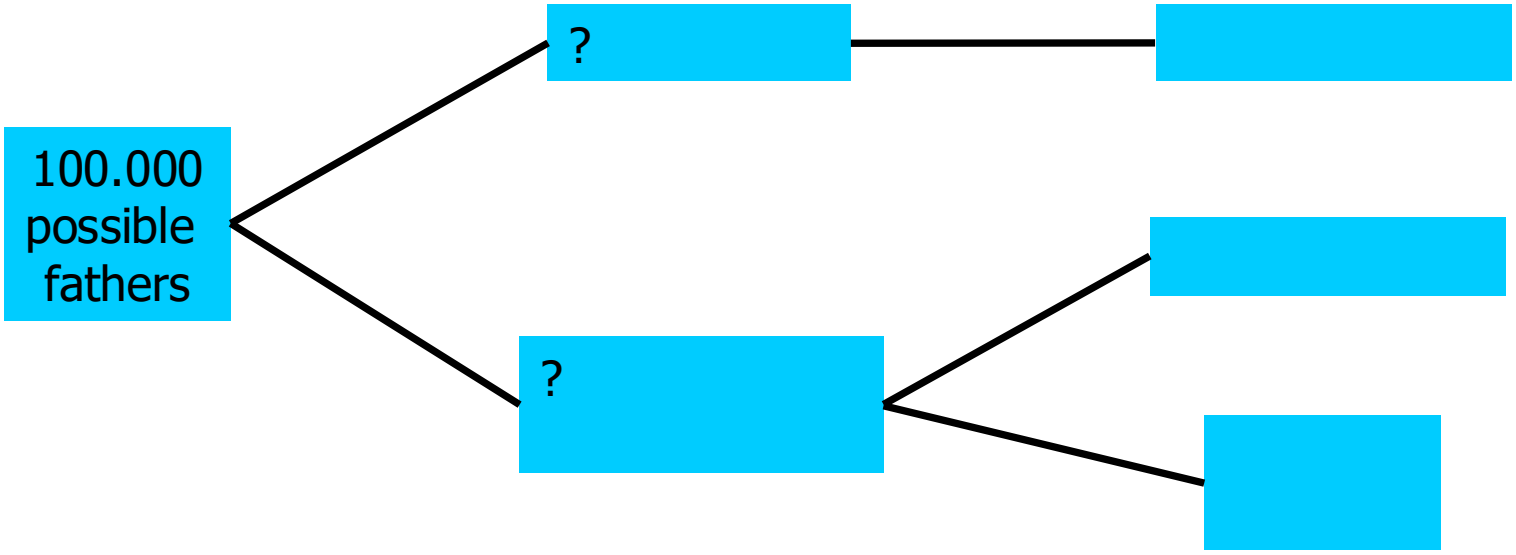


Fatherhood test in tree form

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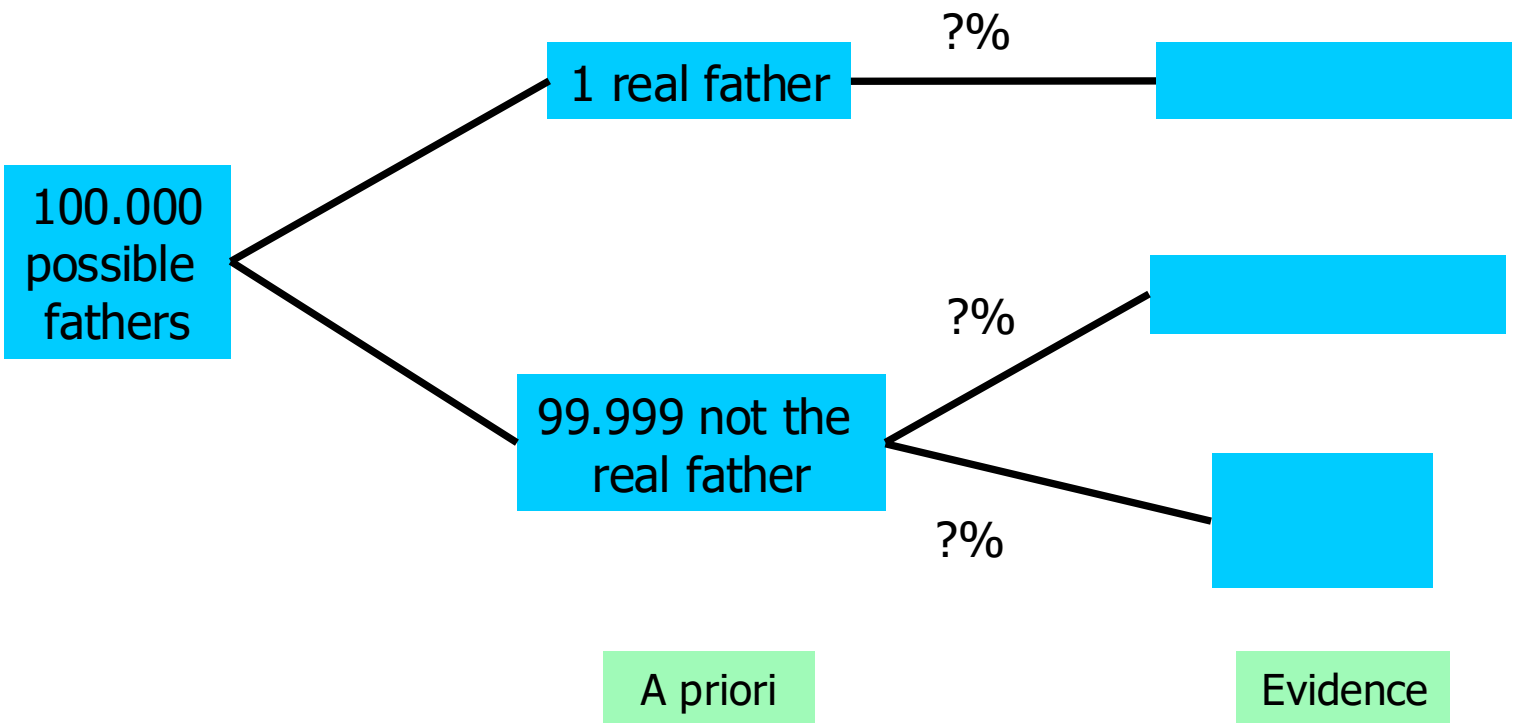


Fatherhood test in tree form

The probability of a negative test given that the person is not the father, is ~~more than~~ 99,99%

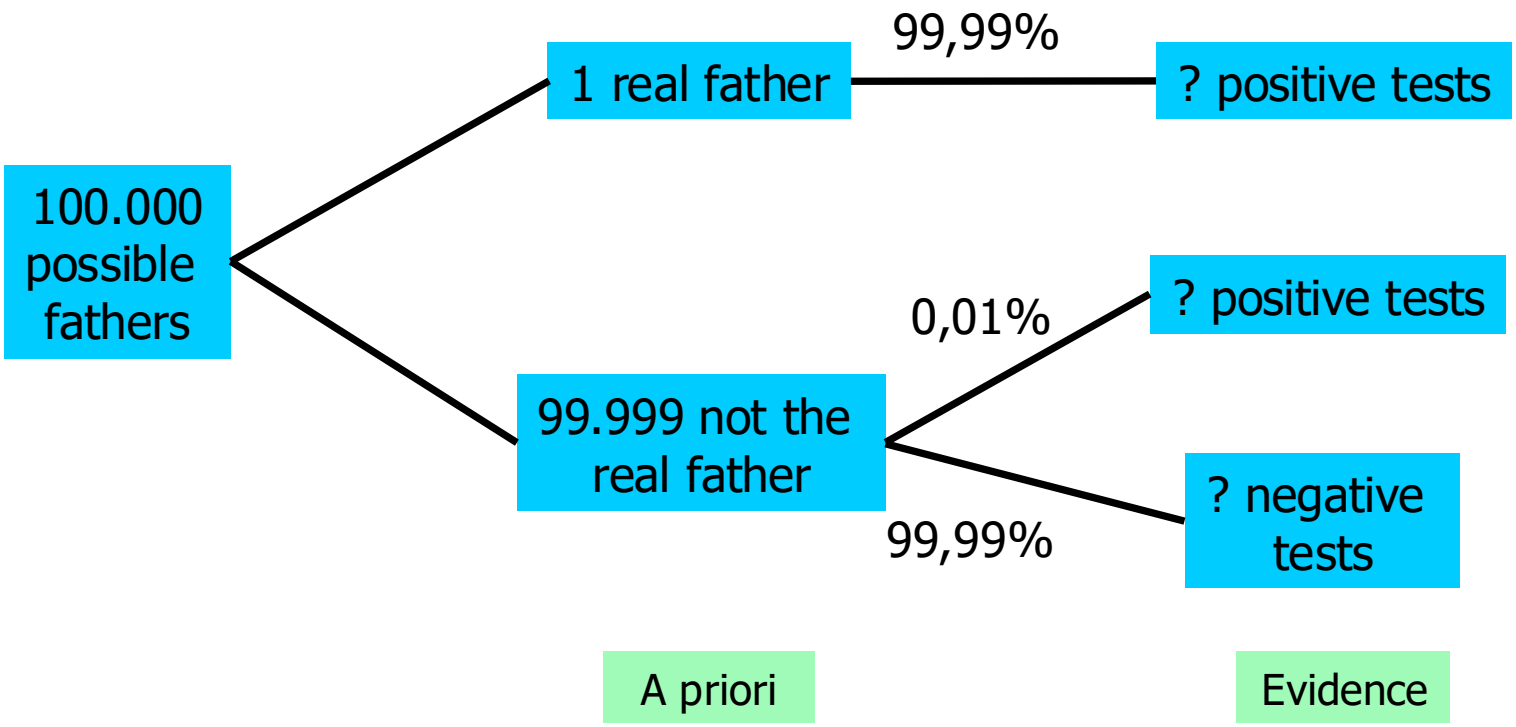
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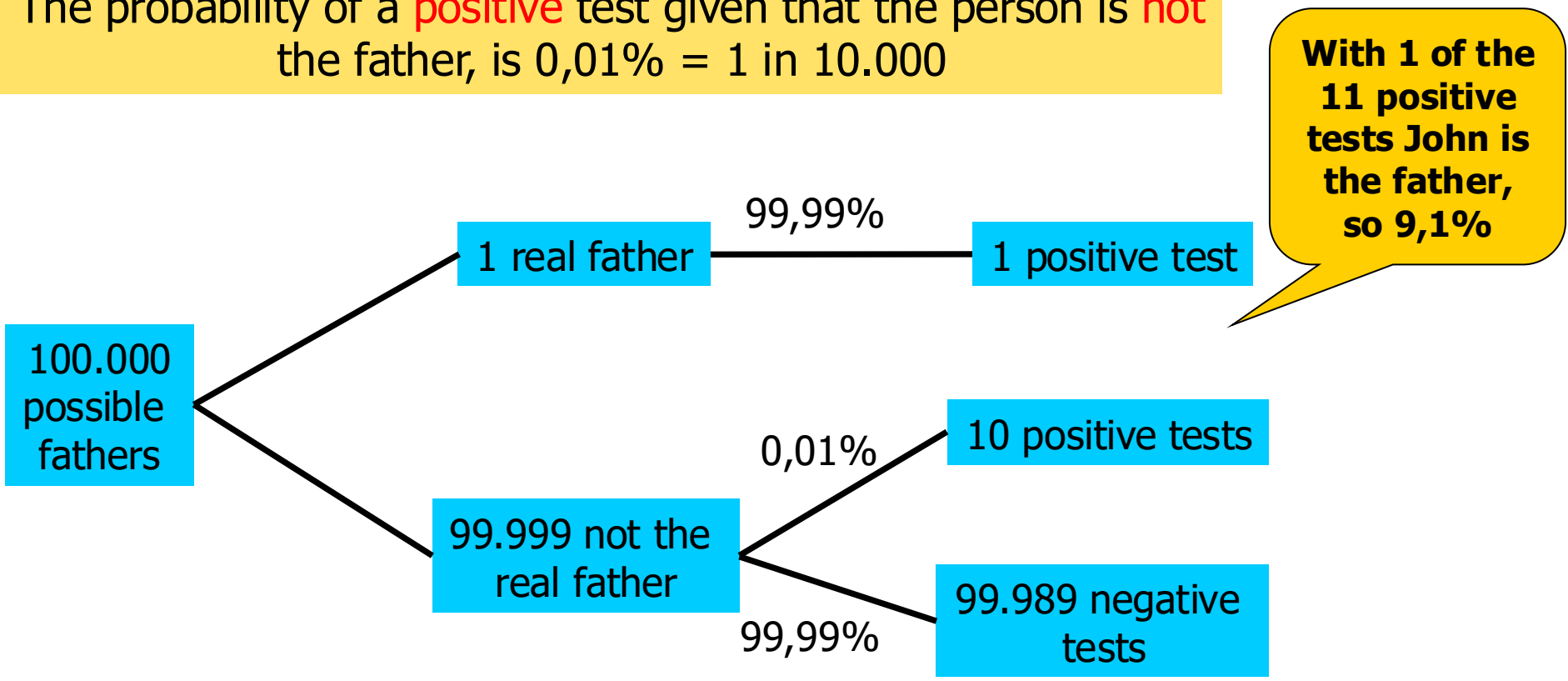


Fatherhood test in tree form

The probability of a negative test given that the person is not the father, is ~~more than~~ 99,99%

So:

The probability of a positive test given that the person is not the father, is 0,01% = 1 in 10.000



Two Forensic Bayesians: N. Fenton & M. Neil (2011)

Unfortunately, while simple examples of Bayes Theorem are easy for statisticians and for mathematically literate people to understand, the same is not true of the general public. ***Indeed, we believe that for many people – and this includes from our own experience highly intelligent barristers, judges and surgeons, any attempt to use Bayes theorem to explain a fallacy is completely hopeless. They simply switch-off at the sight of a formula and fail to follow the argument.***



Bayes' theorem (od

If greater than 1, then E is incriminating evidence,
If less than 1, then E is exculpating evidence,
otherwise E is irrelevant.

The prob of H given E / The prob of not-H given E	=	The prob of E given H / The prob of E given not-H	x	The prob of H / The prob of not-H
Then the computer computes the posterior prob of H given E		Determine or ask an expert to determine the likelihood ratio of E wrt H and not-H		Determine the prior prob of H
Posterior odds	=	Likelihood ratio	x	Prior odds



Taxi with Bayes

$$\frac{\text{Prob of B given "B"}}{\text{Prob of G given "B"}} = \frac{\text{Prob of "B" given B}}{\text{Prob of "B" given G}} \times \frac{\text{Prob of B}}{\text{Prob of G}}$$

$$??? = \frac{???}{???} \times \frac{???}{???}$$

85 Green and 15 Blue taxis
Prob of "Blue" given Blue = 0.8
Prob of "Green" given Green = 0.8



Taxi with Bayes

$$\frac{\text{Prob of B given "B"}}{\text{Prob of G given "B"}} = \frac{\text{Prob of "B" given B}}{\text{Prob of "B" given G}} \times \frac{\text{Prob of B}}{\text{Prob of G}}$$

$$??? = \frac{???}{???} \times \frac{0.15}{0.85}$$

85 Green and 15 Blue taxis
Prob of "Blue" given Blue = 0.8
Prob of "Green" given Green = 0.8



Taxi with Bayes

$$\frac{\text{Prob of B given "B"}}{\text{Prob of G given "B"}} = \frac{\text{Prob of "B" given B}}{\text{Prob of "B" given G}} \times \frac{\text{Prob of B}}{\text{Prob of G}}$$

$$??? = \frac{0.8}{0.2} \times \frac{0.15}{0.85}$$

85 Green and 15 Blue taxis
Prob of "Blue" given Blue = 0.8
Prob of "Green" given Green = 0.8



Taxi with Bayes

41,4%

4

0,176

Prob of B given "B" / Prob of G given "B"	=	Prob of "B" given B / Prob of "B" given G	x	Prob of B / Prob of G
0.704	=	0.8 / 0.2	x	0.15 / 0.85

85 Green and 15 Blue taxis
Prob of "Blue" given Blue = 0.8
Prob of "Green" given Green = 0.8



E: match of John DNA with DNA of Mary's child
G: John is het father

Fatherhood test with Bayes

The prob of G given E / The prob of not-G given E	=	The prob of E given G / The prob of E given not-G	x	The prob of G / The prob of not-G
???	=	???	x	1:1.000.000



E: match of John DNA with DNA of Mary's child
G: John is het father

Fatherhood test with Bayes

The prob of G given E / The prob of not-G given E	=	The prob of E given G / The prob of E given not-G	x	The prob of G / The prob of not-G
1/10	=	10.000	x	1:1.000.000

So posterior prob is 9,1%

Random match probability = 1:10.000

1.000.000 potential fathers

Example: Adams

A rape took place in Hemel Hempstead in 1991. In 1993 Denis John Adams was arrested for another offence and a routine check showed that his DNA matched with that of a sample of semen obtained from the victim of the 1991 rape.
The prosecution's forensic expert estimated the random match probability as 1 in 200 million; the defence thought that 1 in 2 million was a better estimate.
Let's go with the defence: the probability of a match given that Adams is not the source is 1 in 2 million

If you think it does, then you commit the 'prosecutor fallacy'

Does **not** imply that the probability that Adams is not the source given a match is 1 in 2 million.
We must also take the base rate into account. But what is it?

'Prosecutor fallacy': inverting conditional probabilities

"The probability of the match given that Adams is not the source is very low"

Is **NOT** the same as

"The probability that Adams is not the source given the match is very low"

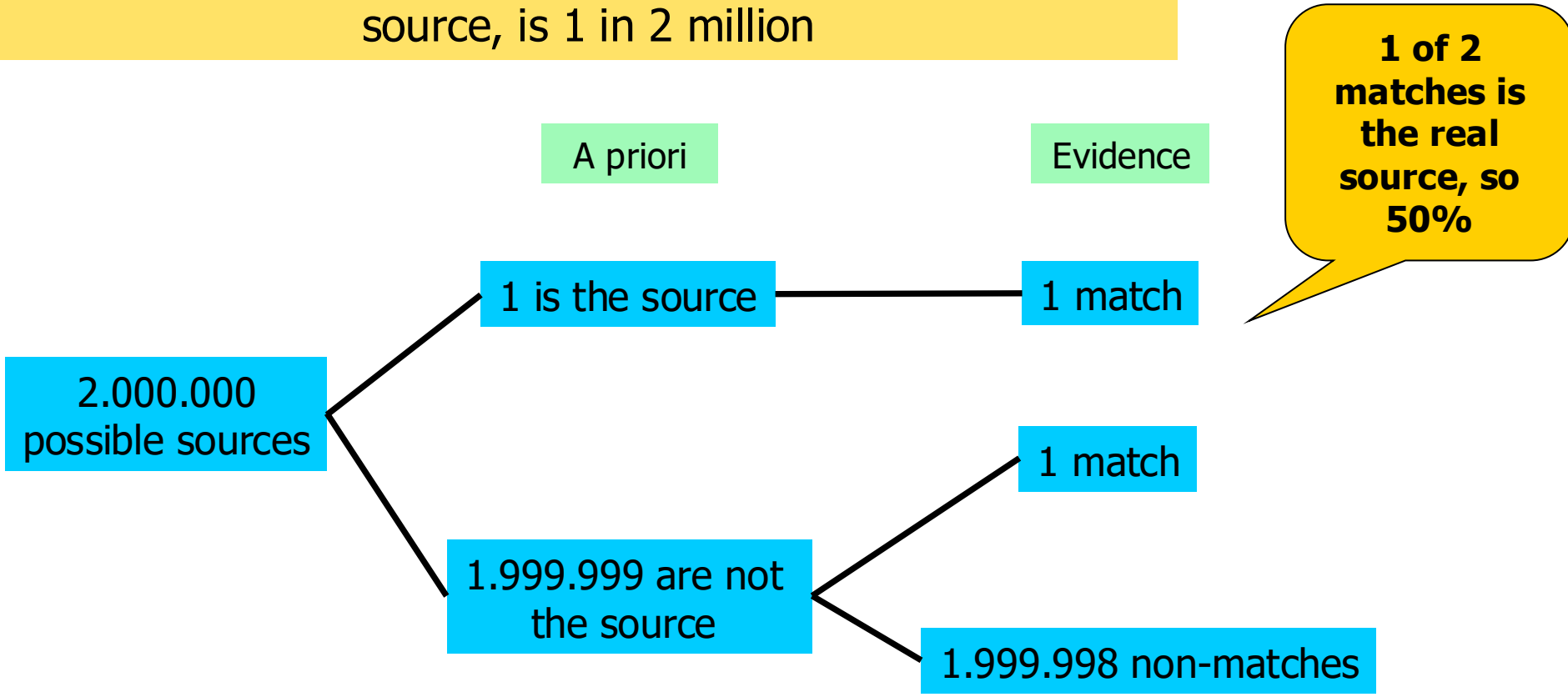
Compare

"The probability that a person is the world's best high jumper given that s/he is tall is very low"

"The probability that a person is tall given that s/he is the world's best high jumper is **NOT** very low"

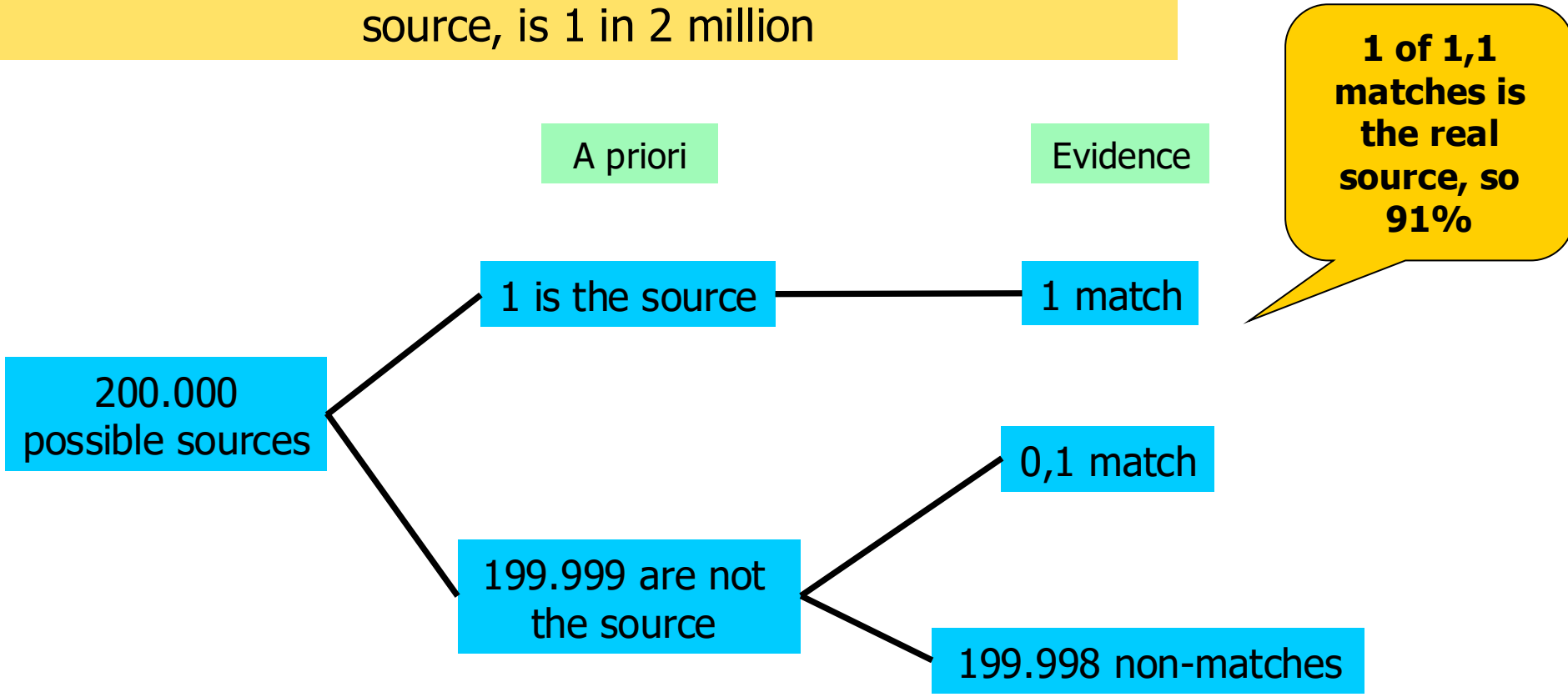
Adams in tree form

The probability of a match given that Adams is not the source, is 1 in 2 million



Adams in tree form

The probability of a match given that Adams is not the source, is 1 in 2 million





E: match of Adams DNA with DNA found at the crime scene
G: Adams is the rapist

Bayesian g

So the prob that Adams is the rapist given the DNA match is 50%

The prob of G given E / The prob of not-G given E	=	The prob of E given G / The prob of E given not-G	x	The prob of G / The prob of not-G
1	=	2,000,000	x	1:2,000,000
		Random match probability = 1:2,000.000		2,000,000 potential sources



E: match of Adams DNA with DNA found at the crime scene
H: Adams is the rapist

Bayesian g

So the prob that Adams is the rapist given the DNA match is 91%

The prob of G given E / The prob of not-G given E	=	The prob of E given G / The prob of E given not-G	x	The prob of G / The prob of not-G
10	=	2,000,000	x	1:200,000
		Random match probability = 1:2,000.000		200,000 potential rapists

Bayes with multiple pieces of evidence in Adams (1)

Prob of Adams' guilt given E1&E2&E3
Prob of Adams' innocence given E1&E2&E3

=

The prior odds of Adams' guilt was **1 in 200.000**

X

The likelihood ratio of the DNA match for Adams' guilt was **2.000.000**

X

The likelihood ratio of the non recognition for Adams' guilt was **??**

X

The likelihood ratio of the girlfriend's alibi for Adams' guilt was **??**

E1 = DNA match,
E2 = non recognition,
E3 = girlfriend's alibi

Bayes with multiple pieces of evidence in Adams (2)

Prob of Adams' guilt given E1&E2&E3
Prob of Adams' innocence given E1&E2&E3

=

The prior odds of Adams' guilt was **1 in 200.000**

1/200.000

x

91%

The likelihood ratio of the DNA match for Adams' guilt was **2.000.000**

10

x

53%

The likelihood ratio of the non recognition for Adams' guilt was **1/9**

10/9

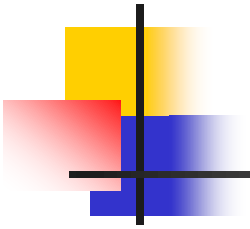
x

36%

The likelihood ratio of the girlfriend's alibi for Adams' guilt was **1/2**

5/9

E1 = DNA match,
E2 = non recognition,
E3 = girlfriend's alibi



Limitations of ‘naive’ Bayes

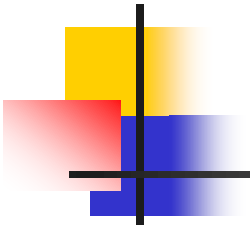
- Repeated updating (or multiplying likelihood ratios) is only justified if the evidence is statistically independent

Example: Sally Clark

In December 1996 Sally Clark's firstborn son died, aged 2½ months, when Sally was alone in the house with him. In January 1998 her second child died, aged 2 months, in similar circumstances.

A paediatrician estimated that the chance that any child from a similar family dies from natural causes is 1 in 8500.
The paediatrician then multiplied 1 in 8500 by itself and concluded that the chance that two children of the same mother die from natural causes is 1 in 73 million.

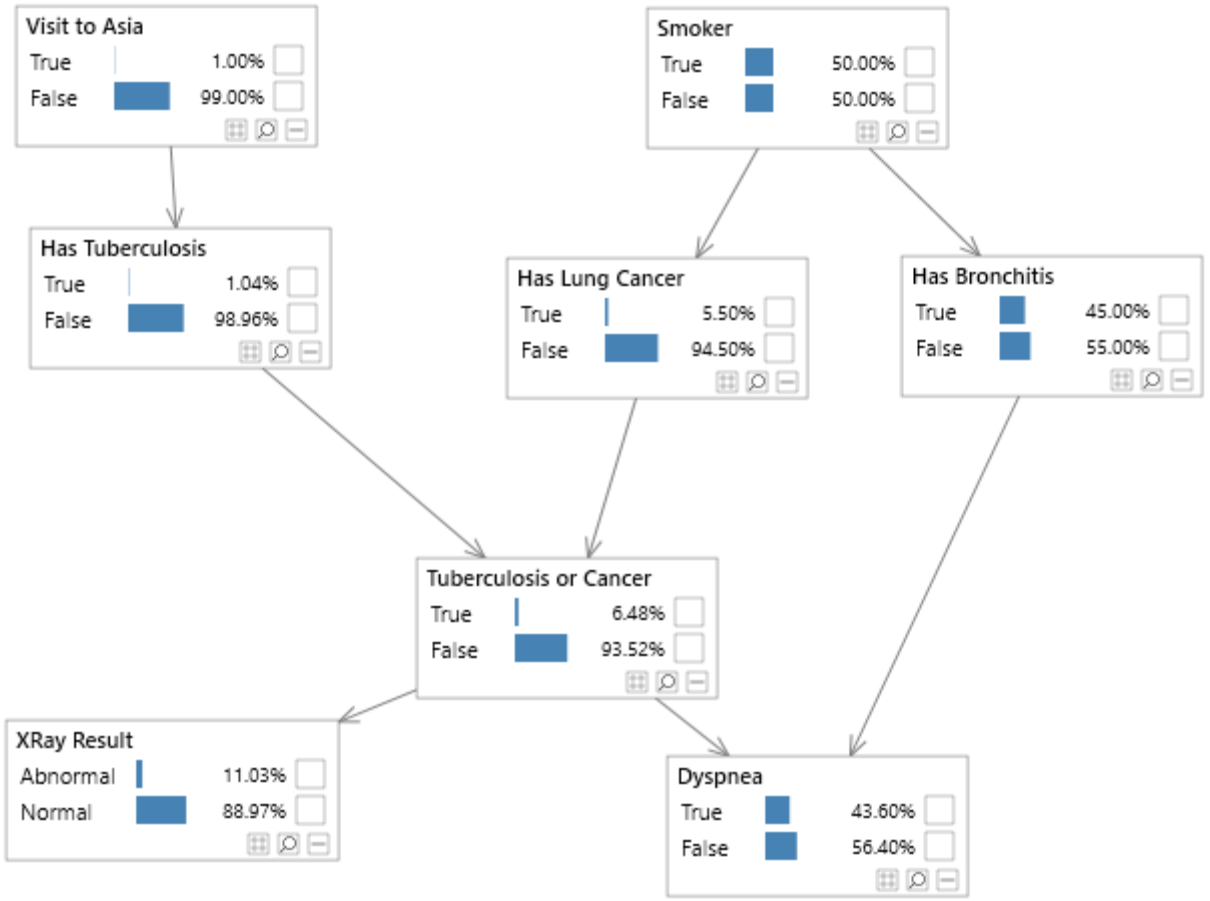
Can you spot the paediatrician's mistake?



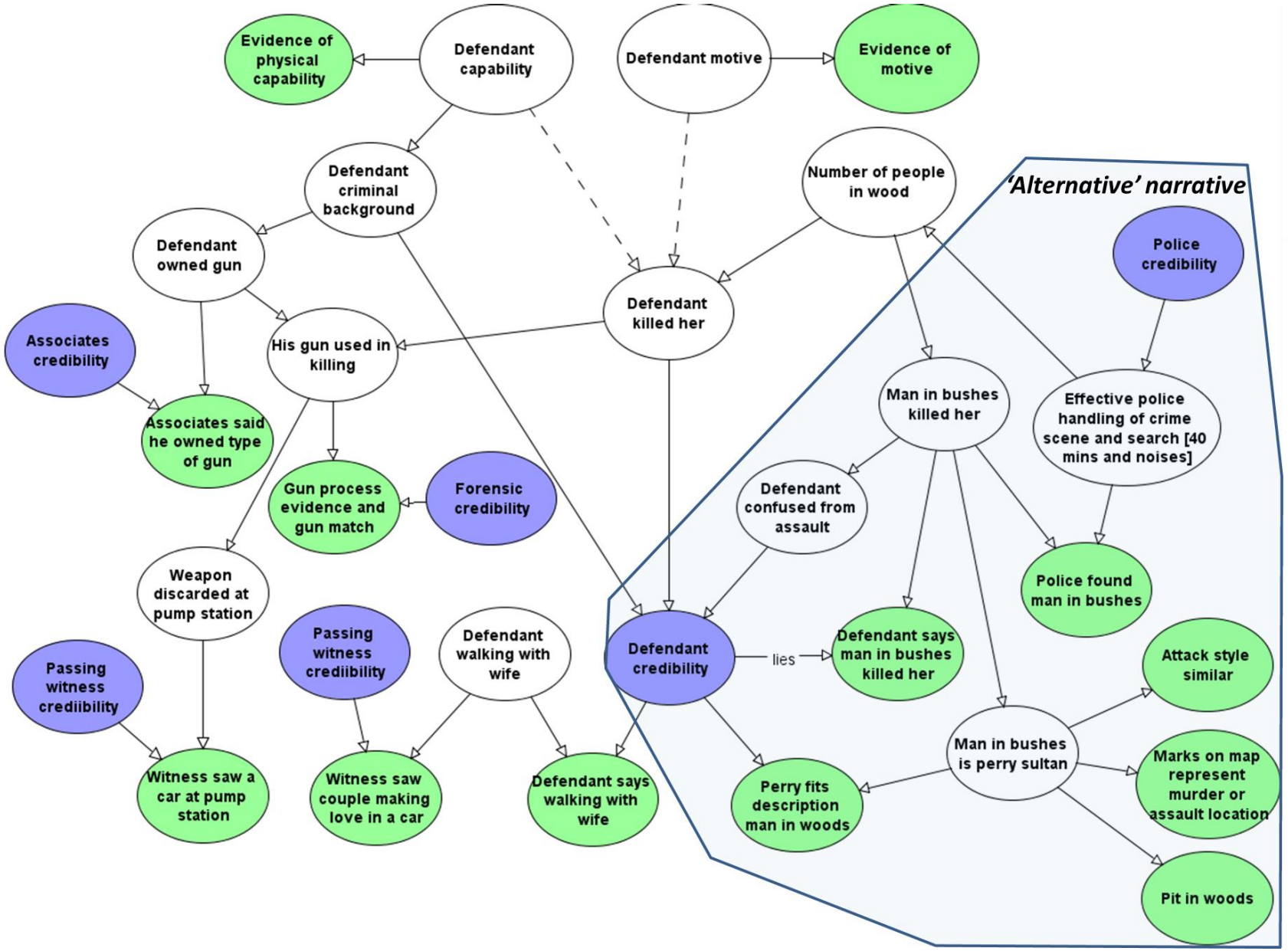
Limitations of 'naive' Bayes

- Repeated updating (or multiplying likelihood ratios) is only justified if the evidence is statistically independent
- Otherwise Bayesian networks are needed(?)
  - Graphically display statistical dependencies
  - Can reduce the number of probabilities to be estimated
  - Their graphical structure can perhaps capture scenarios

Bayesian Network Tools are sensemaking software!

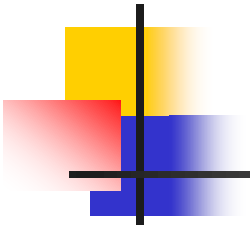


https://www.bayesserver.com/docs/introduction/bayesian-networks



Fenton, N. E., Neil, M., Yet, B., & Lagnado, D. A. (2020). Analyzing the Simonshaven Case using Bayesian Networks. *Topics in Cognitive Science*, 12: 4 (2020), 1092-1114.

Legal proof with Bayes: critical questions



- Can the prior probabilities be reasonably estimated?
- Are the conditional probabilities well-founded?
- To which extent is the evidence statistically independent?
- …



Preventing miscarriages of justice

- Project at University of Groningen, led by Anne Ruth Mackor
- One aim: teach judges to build and understand small BNs of a case
- https://preventingmiscarriagesofjustice.wordpress.com/

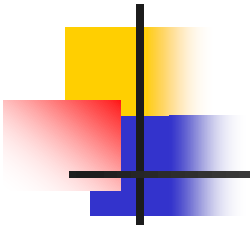
The three approaches: commonalities

- All three approaches can be given mathematical foundations
- All approaches acknowledge that legal proof can never give full certainty
- In all approaches new evidence can change the outcome
- All approaches compare alternatives
- In all approaches generalisations are important

The three approaches: differences (1)

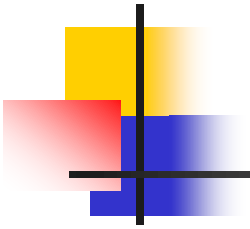
- Argumentation and scenario construction are qualitative, probability theory quantifies uncertainty
  - Do numbers have excessive persuasive force?
  - Or does quantifying uncertainty make judges too cautious?
  - How to compare stories/arguments without numbers?

The three approaches: differences (2)



- Argumentation reasons directly from evidence to hypotheses, the other two reason indirectly from hypotheses to evidence (and then back to the hypotheses)
  - But why does the story-based approach not take priors into account?

My favourite approach (in complex cases)



- Start with constructing alternative scenario's
- Then zoom in on details with argumentation or Bayes

Combinations



- Argumentation + stories
  - Floris Bex (2011), *Arguments, Stories and Criminal Evidence: A Formal Hybrid Theory*. Springer, Dordrecht.



- Argumentation + Bayes
  - Sjoerd Timmer et al., A two-phase method for extracting explanatory arguments from Bayesian networks. *International Journal of Approximate Reasoning* 80 (2017): 475-494.
  - Remi Wieten et al., Information graphs and their use for Bayesian network construction. *International Journal of Approximate Reasoning* 136 (2021): 249-280.



- Stories + Bayes
  - Charlotte Vlek et al., A method for explaining Bayesian networks for legal evidence with scenarios. *Artificial Intelligence and Law* 24 (2016): 285-324.

